

Homework (Habanero)

- Q1.** Solve for x in the equation $2^x = 32$.
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6
- Q2.** If a medicine's potency decreases by a factor of 2 every 3 hours, and its initial potency is 100 mg, what is its potency after 6 hours?
(A) 12.5 mg
(B) 25 mg
(C) 50 mg
(D) 25 mg
(E) 50 mg
- Q3.** A cell doubles its population every 4 hours. If the initial population is 2000 cells, how many cells will be present after 8 hours?
(A) 4000
(B) 8000
(C) 16,000
(D) 32,000
(E) 64,000
- Q4.** The probability P of a patient responding to a new medicine is given by $P = 2^x$, where x is the dosage. If $P = 0.5$, what is the value of x ?
(A) -1
(B) 0
(C) 1
(D) -0.5
(E) 0.5
- Q5.** A disease's spread can be modeled by the equation $y = 3^x$, where y is the number of infected people and x is the number of days. If $y = 243$, what is the value of x ?
(A) 3
(B) 4
(C) 5
(D) 6
(E) 7
- Q6.** A pharmacist needs to package 4 mg, 8 mg, 16 mg, and 32 mg of a medicine. What is the common ratio between consecutive dosages?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 8
- Q7.** The growth of a tumor can be modeled by the equation $V = 2^t$, where V is the volume of the tumor and t is the number of days. If $V = 8$, what is the value of t ?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q8.** A doctor prescribes a 250 mg dosage of a medicine to be taken every 6 hours. If the medicine's potency decreases by a factor of 2 every 3 hours, what is the potency after 9 hours?
(A) 15.625 mg
(B) 31.25 mg
(C) 62.5 mg
(D) 125 mg
(E) 250 mg

Open Ended Questions (FRQ)

- Q9.** A patient's white blood cell count follows the exponential equation $y = 5^x$, where y is the count and x is the number of hours after treatment. Find the count after 2 hours and explain the result.
- Q10.** A new medicine's effectiveness can be modeled by the equation $E = 3^x - 2$, where E is the effectiveness and x is the dosage. Solve for x when $E = 25$ and discuss the implications.

Chef's Tips

- Read each question carefully
- Use a calculator only when necessary
- Show all work and explain your reasoning